

UAVX Flight Test Plan — progressive bench → flight campaign (Spring 2026)

Date: 2026-08-31. Author: Prof. Greg Egan (+ assistant). Companion docs: GCS_Guide.md, Session_Report_InflightAutoTune_Feasibility_Aug29.md (Trace roadmap + Stage-1 capture), wiki/Session_Report_TraceV1_Aug30.md (viewer), wiki/AGENTS.md Trace section. wiki/docs/FW_SIMULATION_REPORT.md / MC_TUNING_REPORT.md for the per-frame sim-predicted response tables.

Rendering: the user turns this into PDF on host “Merlin” via `wiki/docs/md2pdf.sh` (assistant never generates PDFs).

0. Purpose and ground rules

The strategy, stated by the user: **do as much as we can on the bench, then work through the inflight tests progressively as weather permits** (coming into Spring slowly). The feature backlog (Trace V2 layouts, critic C port, CRSF “last-breaths” carrier, in-air tuning) is being **deferred** on purpose until at least one real, trace-verified flight exists — no more features on an unvalidated capture pipeline.

Also: **friends fly** (“all care and no responsibility”) and send logs back for analysis. This plan therefore has two tracks:

- **Track A — benchmark aircraft** (the original/ fleet frames: Ecks_220mm, Ken_450_1165, Ken_Alpha_Test, Ken_LadyBug, Ming_DevEBox, Phoenix, Rok_Quad, S500_1137, WLToys_LadyBug — each has a sim-predicted response to compare against).
- **Track B — guest aircraft** (friend-owned builds). Same capture protocol, but responsibility for the airframe stays with the pilot; we supply the checklist, the analysis, and honest advice. Guest aircraft get validated only to the level the pilot is comfortable with.

Non-negotiables (all flights, both tracks):

1. Propellers clear, crew/observers behind the line, roles named. First battery of the day on every new airframe config is a **no-manoeuver validation** (below).
 2. Fresh, fully-charged flight batteries. Independent BEC for the flight controller where the ESC-integrated 5 V is marginal; NEVER power the FC from a servo rail with a moving airframe.
 3. Failsafe armed and understood per track: RC failsafe presets (NavQual → AltHold+Angle, throttle idle), RTH configured if WP nav is enabled on that frame. Confirm the FS values land as expected on the bench (GCS readback) before the first flight.
 4. The GCS raw log is **always-on** from Connect to Disconnect (per AGENTS: YYYYMMDD_HHMMSS_<Airframe>.rawlog). A flight without a .rawlog did not happen.
 5. Trace captures (ch8) are for **measurement only** — the FC never writes tune changes in the air (write-on-disarm + pilot confirmation only; the in-air APPLY tier is not commissioned).
 6. **Abort criteria at any moment:** F.LQCrI/LQCrI (link), unexpected ExcessLift beeper outside an intended climb, oscillation you did not command, rapid cell sag, smoke/burn smell, props-stopping in a way you did not command. Land immediately; the GCS kept the log.
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1. Benchmark gates — everything that must pass BEFORE any motor turns

Each gate that fails sends us back to the bench. No gate is skipped.

G	Gate	Pass criterion	Where captured
G1	Firmware build	fc_build.py all 6 targets clean; .bin flashed to the test FC; revision label confirms the flashed build on connect	FC + GCS
G2	Param .af round-trip	Load the frame .af, Write, reconnect (tag 72 apply) → GCS readback matches PARAM_DEFAULTS; AirframeName persists (Flash: label + tag-63)	GCS Params
G3	RC mapping + failsafe	Ch6 NavQual/CBW mapping, ch8 → Trace; stick centre values; FS values confirmed present (SBus fail flag or timeout injection)	RC tab readback
G4	Trace capture — anechoic bench	ch8 while State != elnFlight must not open a capture; simulated in-flight settle-step path exercised if the emulator allows; on-disarm TraceCommit writes CAPTURE sector only when armed	FC + GCS Dump
G5	Trace dump + viewer	A committed snapshot (or a synthetic .bin) opens in the viewer; strips render on host; parse_trace round-trip clean	GCS Dump / File → Open Trace Dump
G6	D8 link / cells / vario	(if softserial D8 used) all cells arrive in the 1 s burst; vario scale plausible; no FIFO overrun warning; ~25% softserial utilisation	TX + rawlog
G7	USB-link resilience	Pull/re-plug USB while connected → \${telem} line + reconnect ≤ 5 s; one rawlog per session, named with airframe	GCS
G8	AltHold/level on bench	AltHold engages only when configured and armed; no spur on arm; ExLift lamp clear	GCS flag bytes

Depends on weather: the G4/G5 bench item can be done after a rain-out; G1-G3, G6-G8 are indoor/porch jobs. First Spring flyable days should be spent on the **two easy flights**, not the diagnostic ones.

2. Progressive flight sequence (Track A — benchmark frames)

Each flight N is a *level*; you do N+1 only after N passed fully. Every flight has: preconditions → procedure → capture → pass criteria → record line.

Flight 1 — “No-manoeuve validation” (5-7 min)

Preconditions: G1-G3 pass. Wind < 12 km/h, clear field, no other model in the box. All axes at default. GPS position+fix (if nav enabled). Arm + disarm repeatedly on the bench until the airframe behaviour is boring.

Procedure: Take off to a low hover (MR) / light hand-launch & climb (FW), 20-30 m. **Fly a gentle racetrack only.** No rate steps, no AltHold, no nav. Land. Do not arm again.

Capture: rawlog (whole flight). GCS screenshot with the tag-13/14 numbers at end.

Pass criteria: - Link held for the whole battery (=3 min minimum), no LQCrit, no USB reconnects on the GCS side. - Temperatures sane (ESC/motor hand-<45 °C; SnB reasonable for the craft). - No IWDG reset / unexpected FC behaviour; beeper only where opinion has a reason. - rawlog one file, named ..._<Airframe>.rawlog, replays clean (Ctrl+R) and ends on a disconnect. - (MR) rotor tips visibly stable; no persistent oscillation.

Record line: F1 <date> <craft> <s/n> PASS/FAIL <notes> in the log table (§5).

Flight 2 — Altitude hold flight (5-7 min)

Preconditions: F1 PASS. AltHold configured on this frame; pot/sw mapping per arming rules (rc.c: switch arming → AltHold always on; Tx arming → AltHold > 40 %).

Procedure: Climb to ~30 m. Engage AltHold. Cruise at constant throttle in hold; then give a definite stick step (climb) and release — watch the ROC return to < 0.5 m/s over ~2 s and the altimeter stop. One descent step the same way. Fly home, land.

Capture: rawlog; if this is the very first AltHold flight, also bench-note AltHoldThrComp/DesiredThrottle convergence from the tag-13 stream (the TrackCruiseThrottle 2 %/s feed-forward tracking is visible there).

Pass criteria: - Hold holds: altitude within a few metres at constant power; no porpoise. - Climb/descent steps return to hold cleanly (no 2 s + overshoot, no oscillation). - ExcessLift never fires during a normal cruise (only during an intended climb on soaring-capable frames, and never by accident). - No wind-up stall: after a long forced climb the throttle returns to the previous hold trim without a dip.

Record line: F2 <date> <craft> PASS/FAIL: hold<±m> climb/sink settle<Xs>

Flight 3 — Trace rate probes (measurement flight)

Preconditions: F1/F2 PASS. G4/G5 PASS on the bench. Test axis = Roll (v1 probe). One test per ch8 edge; do not re-arm during a controlled manoeuvre.

Procedure: Steady, level, quiet air. Arm ch8 (OFF→ON) at altitude; FC settles (rates < 0.10 rad/s ~100 ms), opens capture, runs 100 ms pre-roll, then steps $0.25 \times R_{\text{Max}}$ into the Roll rate setpoint for the rest of the window; release ch8 after ~1 s. Repeat 2-3 times in separate sustained-but-settled windows. Land. Dump each capture to the GCS viewer (Dump button → viewer).

Capture: rawlog (whole flight) + each trace dump saved. This is the FIRST validated airborne response data (“the foundation of the whole feature stack”).

Pass criteria: - Capture opens only on the ch8 edge in flight; settle gate holds; the step appears in Desired and Rate responds. - **Compare the measured rise/settle/overshoot/zero-crossings against the sim-predicted values for that frame** (wiki/docs/PID_Simulation_Construction the critic code is the same on both sides). Within ~±30 % and same FAIL set → **the sim plant is validated, the whole tuning story opens.** - If the measured response is cleanly

different (no oscillation, different time constants), that is a *plant-model* finding to record, not a tuning change to make in the air.

Record line: F3 <date> <craft> rise<Xs> ov<X%> settle<Xs> xings<n> vs sim <Δ%>
VALID/DIFF

Flight 4 — Navigation (WP + RTH) flight (only if the frame is nav-capable)

Preconditions: F1-F3 PASS; failsafe RTH confirmed on bench; WP realisation under wind/battery considered (AGENTS TODO — do the ground feasibility check for the day's wind: achievable groundspeed vs. distance + Loiter/RTH reserve BEFORE the leg is sent).

Procedure: Load a short out-and-back (2 WPs, ≤ 200 m, well inside field). Launch, wait for GPS, enable nav. Let it fly the leg; verify turn behaviour and arrival; command RTH from reasonably far; verify home descent and landing gate. Abort-fly manually at least once mid-nav (mode switch) and verify clean handover.

Capture: rawlog (whole flight).

Pass criteria: WP reached within a sensible radius; RTH returns to home and holds; manual takeover is immediate and clean; no ballooning or big cross-track in light wind; battery residual big enough to have loitered 2 min more (energy-reserve evidence).

Record line: F4 <date> <craft> WPA<m> RTH<ok/not> takeover<ok>

Flight 5 — Dive/recovery + VRS protections (commissioned frames only)

Preconditions: F1-F4 PASS for the airframe; DiveRecoverAlt (tag 116), VRSR0C (tag 103), MaxDescentRateDmpS/DescentDelayS **commissioned and tuned together** (AGENTS TODO — never individually). No spectators inside the recovery box.

Procedure: From altitude establish a deliberate descent (power low, nose down on FW / descending on MR); verify the VRS shed happens BEFORE the pull-out threshold; on FW verify the dive-recovery pull-out at DiveRecoverAlt. Recover, climb, repeat once.

Capture: rawlog (whole flight); a trace capture during the dive if you want the rate history of the protect itself.

Pass criteria: Protection engages before any threshold breach; recovery is predictable and never degraded a subsequent manual command; no nuisance trigger in normal flight.

Record line: F5 <date> <craft> VRS_t<Xs before> diveR<ok/no>

3. Track B — guest aircraft (“all care and no responsibility”)

Friends fly **their own** aircraft. We provide the protocol, they fly to it, they send logs. Responsibility for the model rests with the pilot.

What a guest record needs:

- The same .af load-saved and sent, so we can reproduce param state.
- A rawlog: same GCS connect→disconnect. If the guest lacks the GCS, a USB Dump of the trace is preferred but not required — the rawlog is the baseline.
- Flight notes: airframe mass/battery, wind, what was flown (level/alt/mode), anything odd.

Guest progression: recommend they fly the equivalent of F1 and F2 only on their own aircraft (validation + AltHold), and send logs + trace dumps. Do not send guests into F3 rate-probe

territory on their first visit — probing was designed for our benchmark frames; a guest frame's step response is our analysis job, not their flying job.

Analysis we return: a critic comparison of their response against the closest generic frame (Quad, Oct, Spoileron, SkySurfer_Bixler, ...), a PASS/FAIL opinion, and (only on request and only write-on-disarm) a suggested param change. We do not flash friends' aircraft remotely; any .af change is theirs to accept.

4. Log & data custody

Artifact	Produced by	Goes where
.rawlog (always-on)	GCS, per session	~/UAVX/YYYYMMDD_HHMMSS_<Airframe>.rawlog (folder via File → Set Log Folder); copy to logs/<date>/... on the analysis machine
Trace dumps	GCS Dump → viewer / File → Open Trace Dump	save per capture as trace_<date>_<craft>_<axis>.bin alongside the rawlog
.af	GCS Params Save	logs/<date>/... with the rawlog
Sim prediction	tests/test_pid_sim.py	compare on analysis (not per-flight)
Guest logs	sent by friends	dropbox/logs/<guest>/...; keep original filename

Naming convention (strict): YYYYMMDD_HHMMSS_<Airframe>.rawlog (GCS default when the persisted airframe name is present). Guest files keep their GCS-generated name; we never rename the bytes, only the folder.

Analysis workflow after any flight: 1. Replay the rawlog (Ctrl+R) — first sanity: flight re-renders exactly as live. 2. Dump/load the trace capture(s), eyeball the strip. 3. Run the critic metrics + compare vs that frame's sim prediction (this is where measured-vs-sim goes into the running report). 4. Record the flight line in §5 and carry any finding into wiki/Session_Report_<topic>.md.

5. Flight log table (fill per flight)

F	date	craft	sn	wind	battery	duration	mode	result	finding/link
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6. Weather and tempo rules

- **Restart of campaign after a gap > 2 wk:** restart at F1 for the airframe, not where you left off.

- Wind gates per flight: F1/F2 ≤ 12 km/h; F3 (probes) ≤ 8 km/h and no thermal bursts; F4 ≤ 12 km/h with the WP-realisation check; F5 ok to 15 km/h but with altitude margin.
 - A marginal day is a **bench day**: G4/G5, viewer checks, sim/critic re-runs. Do not spend a half-open day on an F5.
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7. Open items that feed the plan (not blocking F1)

- Viewer + critic integration (measure airborne step against the sim on open) — high leverage, bench-buildable before the first trace flight (F3).
- Trace V2 (multiple axes, extended rows), critic C port, CRSF last-breaths — **deferred by design** until F3 produces a validated capture.
- WP-realisation feasibility check (AGENTS TODO) — required only before F4. ““